Estimating Cosmological Parameters from Galaxy Power Spectrum

SIMULATION-BASED INFERENCE

Lecturer:

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1 Introduction

The primary scientific objective of modern cosmology requires scientists to establish the exact values of fundamental universe parameters. The project investigates three fundamental parameters consisting of the Hubble constant (H_0) to measure current expansion rates and matter density (Ω_m) to determine matter content and spectral index (n_s) to understand initial density perturbation characteristics. The parameters function as fundamental components of the present-day standard cosmological model.

The matter power spectrum $P(k)$ represents the most crucial observable to constrain these parameters. The power spectrum of the matter distribution in the universe provides statistical information about the cosmic matter distribution, while its shape strongly depends on background cosmology (the overall expansion history and matter–energy content of the universe, without small-scale fluctuations). The basic cosmological parameters directly influence all aspects of the power spectrum, including its amplitude and slope and its characteristic features. The power spectrum measurement at high accuracy provides an efficient method to determine the best values that describe the universe.

This research presents a computational framework to determine these parameters through simulation-based inference (SBI), which works effectively for complex models with incomputable likelihoods. The *BayesFlow* library uses a neural network to learn the posterior distribution of the parameters $\theta = \{H_0, \Omega_m, n_s\}$. The Relax priors method is used to achieve higher accuracy during training (Charnock, Lavaux, and Wandelt 2018). The training process takes place on synthetic data that use $CAMB$ to generate power spectra, and Gaussian noise is then added to mimic observational data. The document explains the entire process, from data generation through model verification to obtaining the final results. Code link: Github repo

1.1 Motivation

Simulation-Based Inference (SBI) extends the Bayesian approach, providing a means of parameter estimation when direct likelihoods cannot be obtained. Classic Bayesian methodologies work well when there are data from the past or well-defined priors, but they do not work well in fields such as cosmology, where systems are extremely intricate, high-dimensional, and analytically intractable. Inferring major cosmological parameters, the Hubble constant (H_0), matter density (Ω_m), and spectral index (n_s) from the galaxy matter power spectrum, is specifically demanding due to physical interactions that render closed-form solutions unfeasible or impossible.

The challenges can be addressed through the use of likelihood-free inference techniques with neural networks as described in (Alsing et al. 2019; Lueckmann et al. 2018). The

BayesFlow frameworks eliminate the need for likelihood optimization during posterior estimation by using simulated data instead of analytical equations. This paper presents a hybrid convolutional–recurrent network in BayesFlow to summarize multivariate time-series data and to extract long-range dependencies of cosmological parameters such as H_0, Ω_m and n_s for efficient and precise inference.

1.2 Problem Statement

Inferring the matter power spectrum $P(k)$ of cosmological parameters $\{H_0, \Omega_m, n_s\}$ is one of the biggest challenges in cosmology, given the dependence of data, high dimensionality, and the inaccessibility of an analytical likelihood function for these parameters. In the absence of tractable likelihoods, standard Bayesian inference falters, which leaves parameter estimation uncertain and restricts insights into the expansion and structure of the universe (Osato et al. 2023).

To address this issue, the current project uses the BayesFlow framework that leverages deep neural networks to carry out likelihood-free inference. Instead of explicitly estimating the likelihood $P(D|\theta)$, BayesFlow directly estimates the posterior $P(\theta|\mathrm{D})$ using amortized Bayesian inference, which is a scalable method for inferring cosmological parameters from complex simulated data sets (Gómez-Vargas et al. 2024).

1.3 Modeling idea

The work shows cosmological parameter estimation through simulation-based likelihood-free inference methods. The BayesFlow model connects simulated data to parameter estimation in likelihood-free settings by avoiding the calculation of the matter power spectrum likelihood which is computationally intractable. The core idea involves using a neural network to determine cosmological parameters $\{H_0, \Omega_m, n_s\}$ by training it to reverse-map from simulated matter power spectrum data.

First, a $TimeSeriesNetwork$ is used as the summary network. This choice is specifically motivated by the sequential arrangement of data in the power spectrum $P(k)$. Second, for the inference network, the *CouplingFlow* architecture is deployed which initializes an invertible flow-based model with a sequence of transformations. This network is deployed because it can draw new samples faster from the learned posterior distribution (Kingma and Dhariwal 2018; Zhang et al. 2023). The combination of these two networks offers scalable means of modeling the fundamental cosmological parameters, solving the inference problem, and providing better results for the posterior distribution $P(\theta|D)$.

2 Data

The matter power spectrum, denoted as $P(k)$, is commonly employed to describe the distribution of matter in the universe using Fourier space. At a given wave number k, which is the inverse of the length scale, the power spectrum shows the variance of density fluctuations. In linear theory, Fourier modes develop on their own, and the power spectrum is enough to describe the density field on large scales. In mathematical terms, $P(k)$ is the Fourier transform of the density of the two–point correlation function of the matter. The general shape of the matter power spectrum increases with k on very large scales, reaches its peak near the scale of matter–radiation equality, and then decreases toward smaller scales. The peak occurs at $k_{\mathrm{eq}} \approx 2 \times 10^{-2}\,\mathrm{h\,Mpc^{-1}}$ (Bahr-Kalus et al. 2023).

2.1 Data source

This project demonstrates how the power spectrum depends on cosmological parameters. By simulating the non-linear matter power spectrum using the $CAMB$ package. $CAMB$'s interpolator function returns a 2-D spline that evaluates $P(k)$ as a function of both the redshift and the wavenumber (CAMB python package 2025). Here, taking realistic cosmological parameters and adding multiplicative Gaussian noise, a dataset is generated for both training and testing neural posterior estimation.

The true parameters have a physically plausible range of $H_0 \approx 67.4\ km\ s^{-1} Mpc^{-1}$, $\Omega_m \approx 0.315$, $n_s \approx 0.965$. These numbers are close to the Planck 2018-ΛCDM fit (N. Aghanim et al. 2020). The figure 1 shows the noisy matter power spectrum as a function of wavenumbers and redshifts on logarithmic axes.

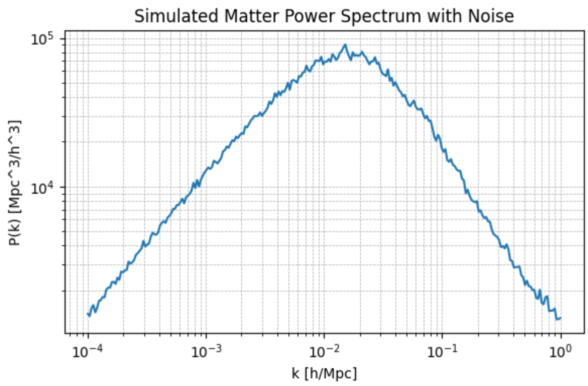

Figure 1: True simulated matter power spectrum $P(k)$ with 5 % multiplicative noise

3 Statistical model (simulator)

BayesFlow manages a generative model as more than just a combination of a prior distribution (encoding beliefs about the parameters before observing data) and a data *simulator* (a likelihood function, often implicit, that generates data given parameters). The user guide shows that a generative model can include additional context assumptions that may be amortized at any time to make inference more flexible. Thus, inference for real-world data is easy. In this project, by default, *SequentialSimulator* is used, which combines multiple simulators into one, sequentially (S. Radev et al. 2023; Stefan T. Radev et al. 2023).

3.1 Explicit proper priors

Proper priors ensure that the resulting posterior distributions are normalizable and avoid any issues. In BayesFlow, priors are defined by arbitrary sampling code—as long as each call returns a valid draw from a probability distribution. The prior function draws cosmological parameters from explicit proper distributions. The Hubble constant H_0 is sampled uniformly. The matter density Ω_m is resampled from a truncated normal distribution. The spectral index n_s comes from a gamma distribution, see table 1. In this way, the priors are broad enough to allow the neural network to learn the sensitivity to deviations. Relaxing prior helps the network sample select from broader, less informative distributions during training, which improves robustness and reduces over-confidence (Charnock, Lavaux, and Wandelt 2018).

Parameter	Prior distribution	Parameters	Range
H_0	Uniform	$a = 55,\ b = 75$	$[55, 75]$
Ω_m	Normal	$\mu = 0.315,\ \sigma = 0.035$	$[0.25, 0.40]$
n_s	Gamma	$\alpha = 15,\ \beta = \frac{1}{15}$	$[0.3, 1.7]$

Table 1: Prior distributions on cosmological parameters used in simulation

Figure 2 visualizes independent draws from the prior over $\theta = \{H_0, \Omega_m, n_s\}$ used throughout simulation-based training. The diagonal panels exhibit the specified marginals, specifically these: $H_0 \sim \mathrm{U}(55, 75)$; $\Omega_m \sim \mathcal{N}(0.315, 0.035^2)$ truncated to [0.25,0.40] (bell-shaped with visible truncation); $n_s \sim \mathrm{Gamma}(\alpha = 15,\ \beta = \frac{1}{15})$ (right-skewed, peaking near 1). The off-diagonal scatter or KDE panels exhibit rectangular supports and do not reveal visible trends that are consistent with parameters existing as independent. Furthermore, each axis uses its own scale to match the support of that parameter, so the ranges differ across H_0, Ω_m, n_s. This diagnostics confirms that the priors place the mass in physically reasonable regions. It additionally verifies that the priors do not unduly constrain the simulation space, being an explicit step in a principled Bayesian workflow before fitting and posterior diagnostics (Betancourt 2020b).

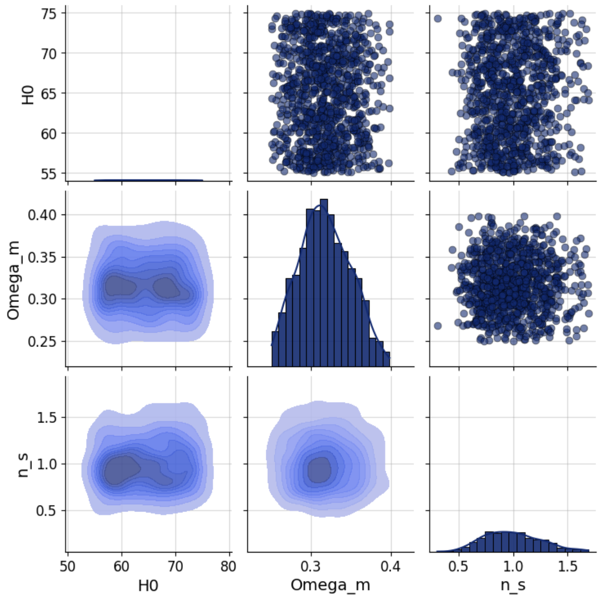

Figure 2: Prior diagnostic pair-plot for $\theta = \{H_0, \Omega_m, n_s\}$

3.2 Data generation

In this section, data sets are generated synthetically by changing a prior likelihood *simulator* with a BayesFlow *adapter*. Each draws samples from broad priors and evaluates a nonlinear matter power spectrum $P(k)$ through *CAMB* on a fixed log space grid $k \in [10^{-4},\ 1]$ with $N_k = 256$ (the discrete wavenumber points k at which $P(k)$ is evaluated). The k-grid is sampled in $N_k = 256$ equally spaced points in logarithmic k.

The BayesFlow `adapter` is a collection of transforms that map the `simulator` output into a form suitable for neural training and inference. It takes the raw parameters (H_0, Ω_m, n_s), concatenates and log-transforms them to form `inference_variables` in $\mathbb{R}^3$; meanwhile, $P(k)$ is mapped to `summary_variable` and reshaped to $(256, 1)$. The resulting dataset comprises a summary tensor of shape $N \times 256 \times 1$ (power-spectrum samples) and an inference tensor of shape $N \times 3$ (cosmological parameters), where N is the number of samples. This standardization of types and scales prepares the spectrum as a 1-D time-series input for the `TimeSeriesNetwork` encoder. The logarithmic transform further stabilizes training across magnitudes (S. Radev et al. 2023). Because CAMB expects positive real physical densities (ω_b, ω_c), the CDM density is floored as $\omega_c \equiv \Omega_c h^2 = \max(\Omega_m h^2 - \omega_b, 10^{-5})$ to enforce $\omega_c > 0$. Otherwise, prior draws with $\Omega_m h^2 \leq \omega_b$ would yield a non-positive ω_c; see (N. Aghanim et al. 2020; CAMB python package 2025) for parameterization and input requirements.

4 Approximator/Workflow

The approximator consists of: (i) pre-sampling and simulation to produce parameter–data pairs; (ii) a learnable summary network (h_ψ) that embeds the simulated data; (iii) a conditional normalizing-flow inference network (f_ϕ) such that, in combination with a base distribution, defines ($q_\phi(\theta \mid h_\psi(y))$); and (iv) an optimizer which works by minimizing the negative of the conditional log-likelihood and adjusting (ψ, ϕ). During inference, the learned (h_ψ) and (f_ϕ) are evaluated repeatedly on observed (y) data to produce an amortized approximation to the posterior ($p(\theta \mid y)$) without further simulation.

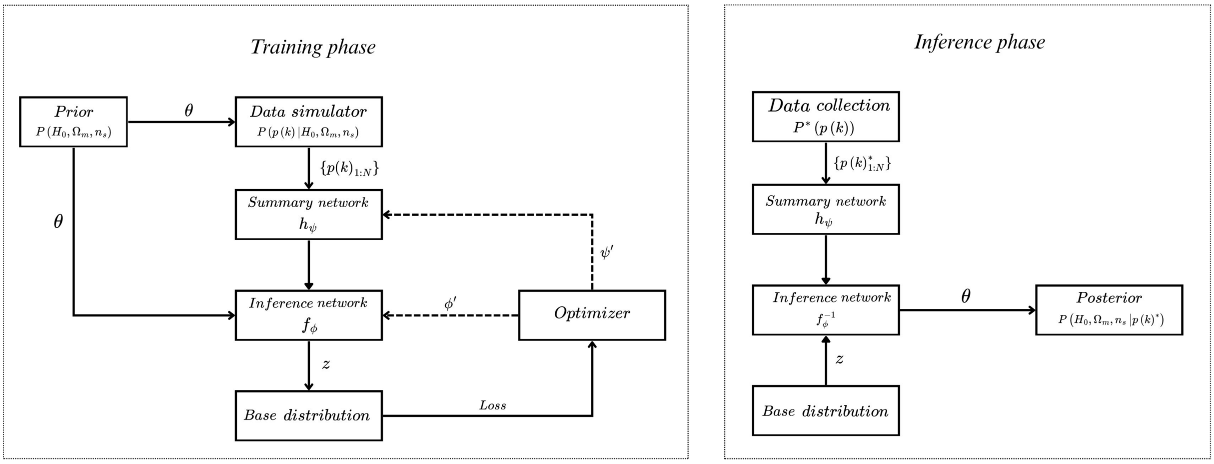

Figure 3: BayesFlow workflow of cosmological parameter (Bürkner and Kucharský 2025)

BayesFlow provides a collection of *approximators* that wrap the inference task and the neural components for executing the task. Here, the application of the `Continuous-Approximator` is employed; it binds the adapter, the summary network h_ψ, and the inference network q_ϕ into a single probabilistic module for the continuous posterior case. At the conceptual level, the approximator achieves the following:

$$q_\phi(\theta \,|\, h_\psi(y)) \;\approx\; p(\theta \mid y)$$

and exposes training, test, and sampling regimes and may control internal standardization of inputs and hyper-parameters. The amortized approximation is precise when the observation (y^{obs}) is drawn from (or close to) the simulation distribution in training; a large distributional shift can decrease posterior accuracy.

4.1 Network architectures

The architecture consists of a summary network ($h_\psi(\cdot)$), which maps the simulated matter power spectrum ($P(k)$) down to a fixed–dimensional latent representation, and an inference network ($q_\phi(\theta \mid h_\psi(y))$), a conditional normalizing flow for providing the posterior over ((H_0, Ω_m, n_s)) given the summary, which enabled facilitate amortized simulation–based Bayesian inference. The learned summary captures relevant temporal/scale structure for conditioning, and the flow sequentially constructs invertible transformations (e.g. coupling blocks) to result in exact likelihoods.

Training optimizes the conditional log–likelihood over sampled parameter–data pairs; during inference, the trained flow embeds a new observation once and outputs posterior samples in one forward pass without further simulation.

4.1.1 Summary network

`TimeSeriesNetwork` - The summary encoder adopts an LSTNet-style design: short-term temporal structures and local inter-series relationships are extracted by a 2D convolutional layer and fed into ReLU and batch normalization; the feature is reshaped and sent into two branches, a regular GRU and a skipped GRU on subsampled slices with skip step (S) (for periodic structure); the outputs of the two branches are fused by a dense layer to generate a summary of a fixed dimension $h_\psi(y)$ which is used to condition the posterior model. This same architecture and reasoning for periodic time series are explained in (Zhang et al. 2023).

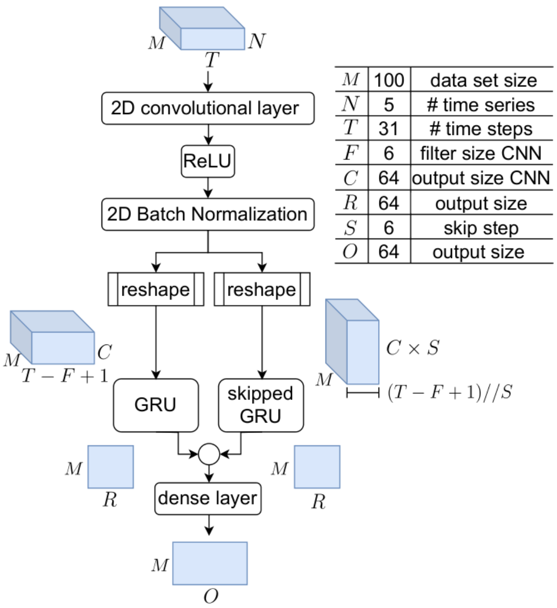

Figure 4: Time-series network architecture (Zhang et al. 2023)

Table 2 lists the parameters used in this project. All values remain at their documented defaults for the training runs reported here.

Table 2: Default configuration of `TimeSeriesNetwork`

Parameter	Value	Description
`summary_dim`	16	Output dimensional of the summary vector.
`filters`	32	Number of filters in each convolutional layer.
`kernel_sizes`	3	Size of the convolutional kernels.
`strides`	1	Stride length for convolutional layers.
`activation`	`"mish"`	Non-linear function in convolutional layers.
`kernel_initializer`	`"glorot_uniform"`	Weight initialization for convolutional kernels.
`recurrent_type`	`"gru"`	Recurrent cell type in the sequence module.
`recurrent_dim`	128	Number of hidden units in the recurrent layer.
`bidirectional`	`True`	Use a bidirectional recurrent layer.
`dropout`	0.05	Dropout rate in the recurrent module.
`skip_steps`	4	Skip length for the skip–recurrent pathway.

4.1.2 Inference network

`Couplingflow` - The posterior $q_\phi(\theta \mid h_\psi(y))$ is modeled by a *conditional normalizing flow* composed of K Glow-style steps (Kingma and Dhariwal 2018). Each step applies (i) *ActNorm* (per-channel affine normalization), (ii) an *invertible* $1{\times}1$ *convolution* (learned channel mixing), and (iii) an *affine coupling* transform. In the coupling layer, the input is split $x = (x_a, x_b)$; a conditioner $g_\phi(x_a, h_\psi)$ outputs scale and shift (u, t), and updates $x'_b = u \odot x_b + t$ while $x'_a = x_a$, yielding a triangular Jacobian with closed-form log-determinant. Stacking these steps as shown in figure 5 defines an invertible map $f_\phi(\cdot; s)$ with exact likelihood via change of variables is given by $\log q_\phi(\theta \mid h_\psi) = \log p_0\Big(f_\phi(\theta; h_\psi)\Big) + \sum_{i=1}^{K} \log|\det J_i|$. At inference, samples are drawn by $z \sim p_0$ and inversion $\theta = f_\phi^{-1}(z; s)$. Table 3 lists the parameters used in this project.

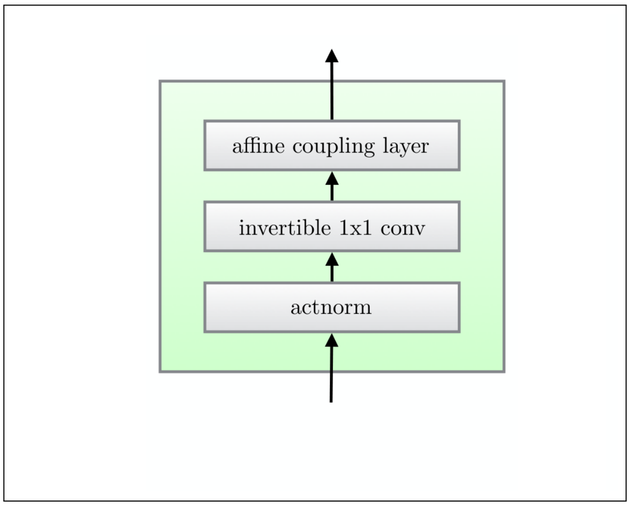

Figure 5: Coupling flow architecture (Kingma and Dhariwal 2018)

Table 3: Configuration of `CouplingFlow`.

Parameter	Value	Description
`subnet`	`"mlp"`	Architecture for the transformation network.
`depth`	6	Number of invertible flow steps (ActNorm $\rightarrow$ permutation $\rightarrow$ coupling).
`transform`	`"affine"`	Type of coupling transform.
`permutation`	`"random"`	Channel mixing between steps.
`use_actnorm`	`True`	Apply ActNorm before each coupling layer.
`base_distribution`	`"normal"`	Latent base distribution used in change of variables.

5 Model Training and Evaluation

The BayesFlow approximator outlined in this work is run in an offline setting. Precomputed simulations are used before training begins, instead of being computed in real time. This approach eliminates the need for constant access to the expensive CAMB simulator and allows for stable convergence across many iterations. Furthermore, offline setup improves reproducibility and consistency during the training process, which is especially crucial in simulation-based inference, where hyper-parameter calibration and performance evaluation are key aspects (S. T. Radev et al. 2020; Cranmer, Brehmer, and Louppe 2020).

5.1 Training budget

The model's training schedule consisted of a total of 150 epochs, utilizing 256 batches as batch size. There were about 40 training batches per epoch and 8 validation batches per epoch. The data constituted 10,000 training samples and 2,000 validation samples. The setup allowed for peak computational effectiveness, delivering the statistical strength required to make proper judgments about performance.

Table 4: Training configuration and number of samples.

Parameter	Value
Training set size	10,000 samples
Validation set size	2,000 samples
Epochs	150
Batch size	256
Training batches per epoch	$\sim$40
Validation batches per epoch	$\sim$8

5.2 Training strategy

In this report, the training and validation sets were *pre-simulated once* and reused across all experiments, ensuring that every configuration was evaluated under identical data conditions. This design removes variability from fresh simulator draws, isolates the effect of architectural and optimizer choices, and improves run-to-run comparability (residual stochasticity from optimization is mitigated via fixed random seeds).

An *amortized offline training* regime is employed: A conditional posterior is learned in a large bank of prior–predictive simulations and then reused for fast inference on new observations. This approach is standard in simulation-based inference for computationally demanding forward models (e.g., cosmological power spectra), where the cost of simulation dominates and amortization pays that cost upfront (Greenberg, Nonnenmacher, and Macke 2019).

5.3 Model configuration

Optimizer and Learning Rate: The optimization procedure adopted in this research uses the AdamW optimizer (Loshchilov and Hutter 2019). AdamW is an improvement over the widely used Adam optimizer, wherein the weight decay component is decoupled from the gradient update mechanism. This design difference prevents the optimizer from causing overly aggressive regularization, thus improving the model's generalization ability in the context of an intricate architecture design. The initial learning rate is 5×10^{-4}, and the weight decay factor is 1×10^{-5} in this case. The learning rate in this project follows the AdamW update rule with decoupled weight decay:

$$\theta_{t+1} = \theta_t - \eta_t \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} - \eta_t \cdot \lambda \, \theta_t,$$

where η_t is the learning rate at step t, $\hat{m}_t$ and $\hat{v}_t$ are the bias-corrected moment estimates of gradients, ϵ is a small stability constant, and λ is the weight decay coefficient. In this project, $\lambda = 1 \times 10^{-5}$ and the initial learning rate is $\eta_0 = 5 \times 10^{-4}$ (Zhou et al. 2024).

The ReduceLROnPlateau policy is adopted to improve convergence by utilizing a learning schedule that is validation-metric-dependent. In particular, in the absence of noticeable improvement in the validation loss across six consecutive epochs, the learning rate is reduced by half (KerasTeam 2025). In this way, the optimizer can make more significant exploratory changes in the earlier epochs of training, with gradually more accurate adjustments in the later epochs. To avoid the learning rate becoming too small, thereby stagnating progress, a minimum threshold of 10^{-6} is set. This adaptive method ensures that the training makes progress without allowing the risk of premature convergence to take hold.

Table 5: Optimizer and learning rate settings.

Setting	Value
Optimizer	AdamW
Initial learning rate	5×10^{-4}
Weight decay	1×10^{-5}
LR schedule	ReduceLROnPlateau
Patience	6 epochs
Reduction factor	0.5
Minimum LR	1×10^{-6}

5.4 Training behavior and observations

In figure 6The learning and validation loss curves both demonstrated a stable decreasing trend in the initial 50 epochs before leveling off. Low variance in these curves indicated that the model was moving toward good generalization rather than overfitting. At the end of the training process, the validation loss had stabilized in the range of about 0.7 to 0.8 nat, indicating the ability of the model to generate properly calibrated posterior estimates of the cosmological parameters.

The excellent performance can be attributed to the combined action of the AdamW optimization method with the adaptive learning rate schedule mechanism. Using the AdamW optimizer improves stability by decoupling weight decay from gradient updates, a property that is especially beneficial for deep-generative networks. At the same time, the validation-based scheduler adjusts the learning rate according to the convergence signals, thus finding the optimal trade-off point between efficiency and robustness. When combined with the offline training system, these design choices follow popular practices in simulation-based inference, thus explaining the outstanding overall performance. (Papamakarios et al. 2021).

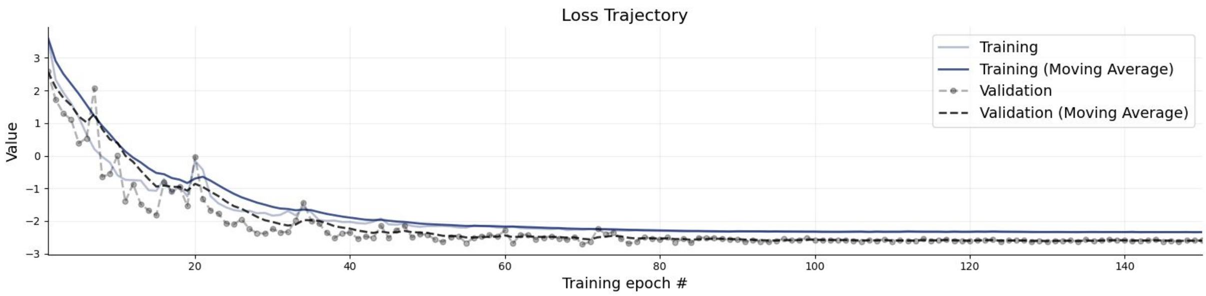

Figure 6: Training and validation loss trajectory

6 Diagnostics

In this section, model performance will be discussed using various diagnostic plots such as posterior calibration, posterior z-score, normalized root mean square error (NRMSE) and posterior contraction for 200 new data sets.

6.1 Posterior contraction vs posterior z score

Following the "Bayesian Eye Chart" (Betancourt 2020a), each test case yields a *posterior z-score* (distance between posterior location and truth in posterior s.d. units) and a *posterior contraction* (shrinkage relative to the prior).The patterns in the scatter diagnose lack of information (low contraction, small $|z|$), prior–likelihood tension (low contraction, large $|z|$), overconfidence/bias (high contraction, large $|z|$), and good fit (high contraction, small $|z|$).

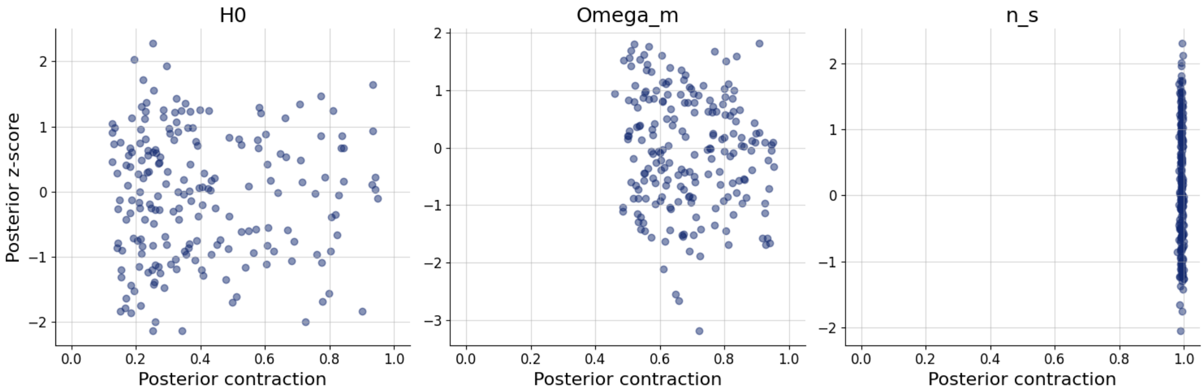

Figure 7: Posterior contraction vs posterior z score plot

H_0: the posterior contraction ranges from low to high (many points around 0.2 to 0.4) with most z-scores in $[-2, 2]$. This pattern shows weak–to–moderate information; Ω_m: the posterior contraction is mostly moderate to high ($\gtrsim 0.5$) and the posterior z-score close to 0, indicating informative and well-centered posteriors with only minor location error. n_s: the posterior contraction clusters with all clustered to 1 1 and the posterior z-score ranging from [-1, 1], indicating a highly informative and well-centered posteriors. Overall: calibration appears *moderate to good*; information is strongest for n_s and Ω_m, while H_0 is least identified. These readings follow diagnostics (Betancourt 2020a, p. 1.3.2).

6.2 Rank Statistics

Calibration was assessed with BayesFlow's *calibration_histogram*, which plots *rank statistics* for each parameter using simulated test pairs and posterior draws. For parameter j

in test case i, with true value $\theta_j^{*(i)}$ and M posterior draws $\{\theta_j^{(i,m)}\}_{m=1}^{M}$, the SBC rank is

$$R_j^{(i)} = \sum_{m=1}^{M} \mathbf{1}\left\{\theta_j^{(i,m)} < \theta_j^{*(i)}\right\},$$

which is *uniform* on $\{0, \ldots, M\}$ when the posterior is calibrated (Talts et al. 2020, Sec. 2).

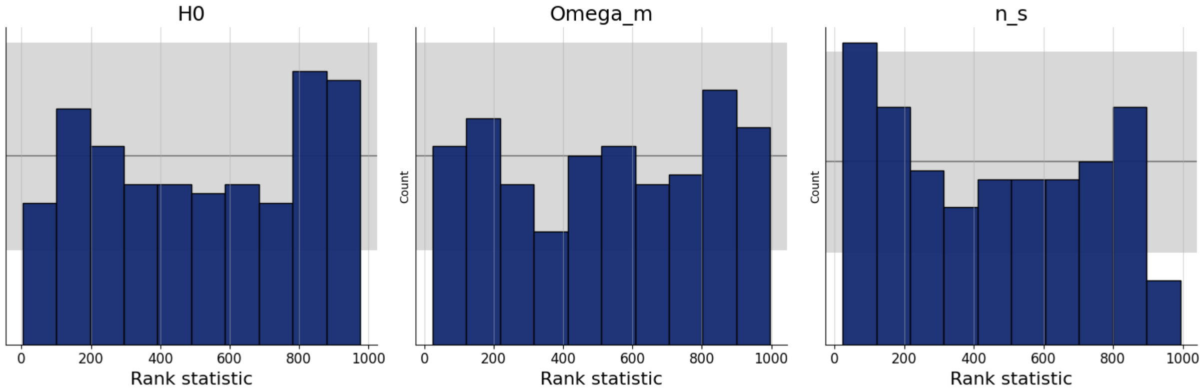

Figure 8: Rank statistics histogram

Under Simulation-Based Calibration, a calibrated posterior has rank histograms that are (approximately) uniform; systematic shapes indicate pathologies. In rank statistic figure 8, H_0 has a moderate peak in the high-rank bins, a pattern of a small location bias where posterior draws are on average below the true value. Ω_m looks widely flat and remains in the binomial uncertainty band, which is consistent with calibration and n_s has the shape of a shallow $\cup$ — a large mass in the extreme bins with a low point near the center, which is consistent with mild under-dispersion (overconfidence). Overall, the parameters are adequately calibrated: the deviations are visible but are retained low in relation to the uncertainty band. (Talts et al. 2020, Sec. 2)

6.3 Normalized root mean square error (NRMSE)

To summarize recovery of point estimates, the normalized RMSE is reported for each parameter:

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{\theta}_i - \theta_i^*)^2}, \qquad \text{NRMSE}_\sigma = \frac{\text{RMSE}}{\text{sd}(\{\theta_i^*\}_{i=1}^{N})}.$$

Here, $\hat{\theta}_i$ is the posterior mean (or median) of the test case i, and θ_i^* is the ground truth. Normalizing by the empirical standard deviation yields a dimensionless metric comparable across parameters (Hyndman and Koehler 2006).

In table 6, n_s is recovered most accurately (≈ 0.025), followed by Ω_m (0.176); H_0 is weakest (0.321). The contraction mirrors this pattern high for n_s (≈ 0.99), moderate–high for Ω_m (≈ 0.66) and lower for H_0 (≈ 0.33). All calibration errors are small and comparable, which, together with the above plots, is compatible with moderate-to-good calibration overall.

Table 6: NRMSE, posterior contraction, and calibration error per parameter.

	H0	Ω_m	n_s
NRMSE	0.321020	0.176362	0.024868
Posterior contraction	0.334115	0.664863	0.992609
Calibration error	0.024474	0.021447	0.034079

7 Inference

After training, the BayesFlow posterior was evaluated on simulated noisy matter power spectra to recover the joint and marginal distributions of the Hubble constant H_0, the matter density Ω_m, and the scalar spectral index n_s. The pairwise posterior plot in Fig. 9 shows unimodal marginals for all parameters; the ground-truth values (red markers / lines) fall within high–density regions of their posteriors, indicating accurate localization of the parameters in these test cases (S. Radev et al. 2023).

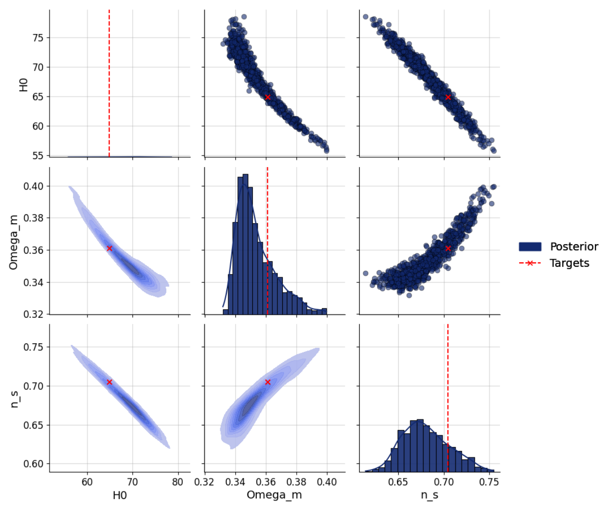

Figure 9: Pairwise posterior distributions of H_0, Ω_m, and n_s.

The off-diagonal panels exhibit physically expected degeneracies: a strong *negative* correlation between H_0 and Ω_m from distance–redshift constraints in ΛCDM, a *positive* correlation between Ω_m and n_s, and a *negative* correlation between H_0 and n_s (Alsing et al. 2019). Marginal widths appear sensible and free of multi-modality; no visual evidence of mode collapse or extreme overconfidence is apparent in this example. Taken together, the pairwise plot is consistent with a well–behaved learned posterior that respects known cosmological parameter couplings while concentrating around the simulated targets.

As expected by the ΛCDM forecast (N. Aghanim et al. 2020), the posterior distributions also had a strong negative correlation between H_0 and Ω_m, illustrating their degeneracy in estimating cosmological distances. The correlations with n_s were weaker, as expected for their role in dictating the form of the matter power spectrum but not its general scale. Marginal distributions had physically sensible widths and forms, without signs of mode collapse or over-confidence, thus indicating robust uncertainty coverage.

In addition to this, credible intervals successfully captured the expected variability due to observational noise and cosmic variance. The results show that BayesFlow successfully retrieves accurate central parameter estimates while producing calibrated and interpretable posterior distributions, further cementing its status as a trustworthy tool for simulation-based inference and a viable option for use on real survey data (Gómez-Vargas et al. 2024).

8 Limitations and potential improvements

Although the current model delivers encouraging results, several limitations must be acknowledged. First, the analysis is restricted to the estimation of only three cosmological parameters. Second, the observational noise in the present set-up is modeled with Gaussian errors of uniform variance. In practice, survey data frequently exhibit scale-dependent and non-Gaussian noise, which cannot be fully addressed within the framework of the current methodology (Zarrouk et al. 2019). Finally, the simulations used for training and validation do not incorporate key astrophysical systematics, including baryonic effects, survey window functions, and other observational complexities. These omissions raise the possibility of bias when transitioning from simulated datasets to real observations, underscoring the need for further investigation and refinement of assumptions in future research. .

To improve the model's generalizability and transferability, three extensions are suggested. First, it is necessary to extend the inference space to cover a wider range of cosmological parameters, such as spatial curvature (Ω_k), cumulative neutrino mass ($\sum m_\nu$) and dark energy parameters (w_0, w_a). Second, it is a good idea to replace simplified Gaussian noise with scale-dependent realistic noise models and incorporate astrophysical systematics, e.g., baryonic effects (Miller et al. 2013). Third, the approach should be tested using survey-grade DESI and Euclid simulations, with benchmarks set against unique constraints from the cosmic microwave background (Franco Abellán et al. 2025). These actions would improve the model's predictive power, calibrate it better under real observational regimes, and increase its overall scientific utility.

9 Conclusion

This report implemented a simulation-based amortized inference pipeline with BayesFlow. The architecture combines a `TimeSeriesNetwork` summary encoder with a conditional `CouplingFlow` to estimate posteriors of (H_0, Ω_m, n_s) from noisy matter power spectra (S. T. Radev et al. 2020; Papamakarios et al. 2021). Offline training on a fixed bank of prior–predictive simulations enabled stable optimization and comparisons between configurations.

Diagnostics indicate posterior behavior *moderate-to-good*. Simulation-Based Calibration rank histograms were broadly compatible with uniformity, and the Bayesian eye chart posterior z-score versus contraction showed informative, well-centered posteriors for Ω_m and n_s with smaller residual issues for H_0. The accuracy of the point estimate mirrored these findings: NRMSE was lowest for n_s, moderate for Ω_m, and highest for H_0. Pairwise plots revealed physically expected parameter couplings, including the familiar negative correlation between H_0 and Ω_m, with ground-truth values lying in high-density posterior regions(N. Aghanim et al. 2020).

Taken together, the results support the suitability of the BayesFlow architecture for cosmological parameter inference from summary spectra: the model captures key degeneracies, yields calibrated uncertainty for most parameters, and recovers accurate point estimates on simulated tests. The remaining gaps, especially the weaker identifiability of H_0—can be addressed by expanding the parameter space, enriching the noise/systematics model, and validating on survey-grade simulations, which are outlined as priorities for future work.(Agarwal and Feldman 2010; Miller et al. 2013)

10 Reflection on own learnings

This project consolidated theoretical and practical skills in simulation-based inference (SBI). The work implemented amortized, likelihood-free Bayesian estimation to recover posteriors from noisy, high-dimensional spectra and developed a clearer understanding of identifiability, parameter degeneracies, and the role of informative summaries (Cranmer, Brehmer, and Louppe 2020). Architecturally, the pipeline combined an Adapter, a `TimeSeriesNetwork` (conv–recurrent) summary encoder, and a conditional `CouplingFlow` (normalizing flow), reinforcing concepts such as change-of-variables, exact log-likelihoods, and conditioning mechanisms(S. T. Radev et al. 2020; Papamakarios et al. 2021; Kingma and Dhariwal 2018).

Evaluation practice improved through targeted diagnostics: Simulation-Based Calibration (rank histograms), the Bayesian eye chart (posterior z-score vs. contraction), Pairwise plots for joint structure, and NRMSE for point-estimate accuracy. These tools strengthened intuition for distinguishing accuracy from calibration and for reading uncertainty under misspecification of the model.

On the engineering side, the work addressed input dimensionality and preprocessing of noisy simulations, stabilized optimization (AdamW with scheduling) and enforced run-to-run comparability via fixed seeds and fixed offline simulation banks. The result is a more robust workflow for training, validating, and interpreting amortized posteriors (Loshchilov and Hutter 2019; KerasTeam 2025; Zhou et al. 2024).

Overall, the project sharpened the link between simulation design, probabilistic modeling, and neural architectures, providing a solid foundation for applying SBI to survey-grade cosmological analyses (N. Aghanim et al. 2020; Osato et al. 2023; Berti, Spinelli, and Viel 2024)..

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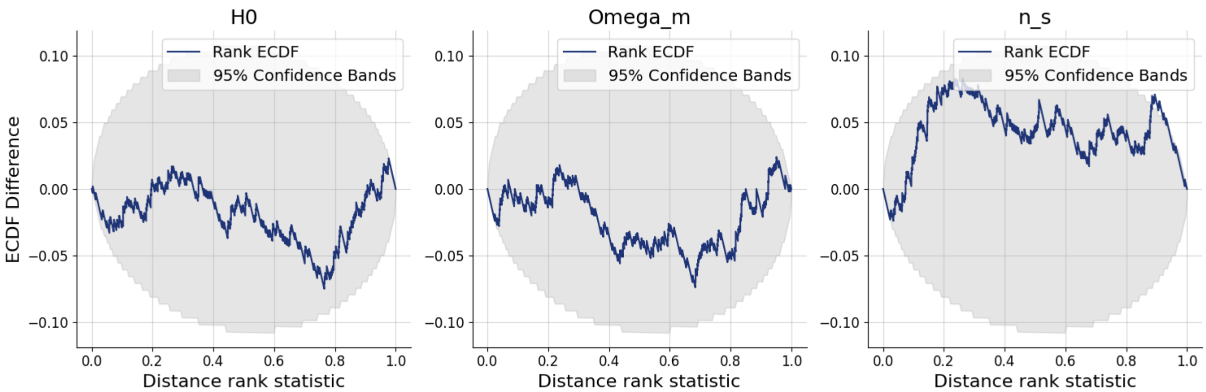
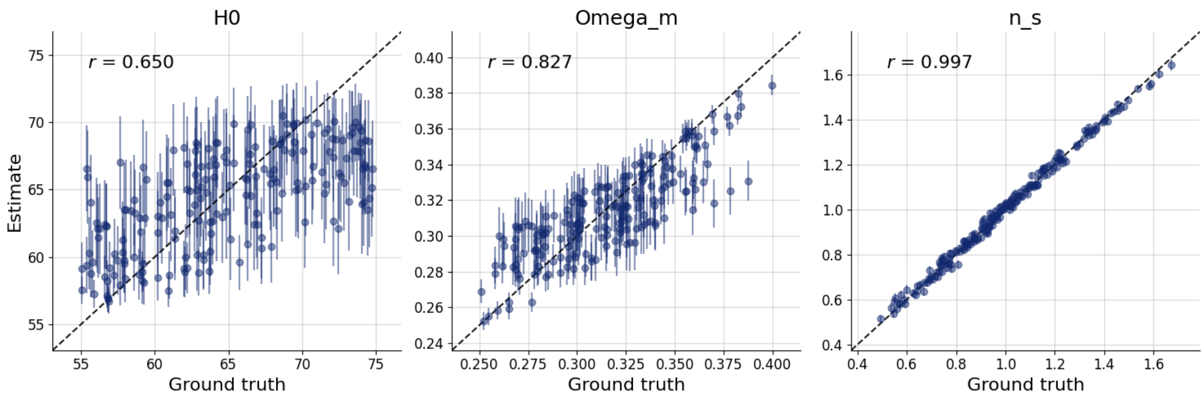
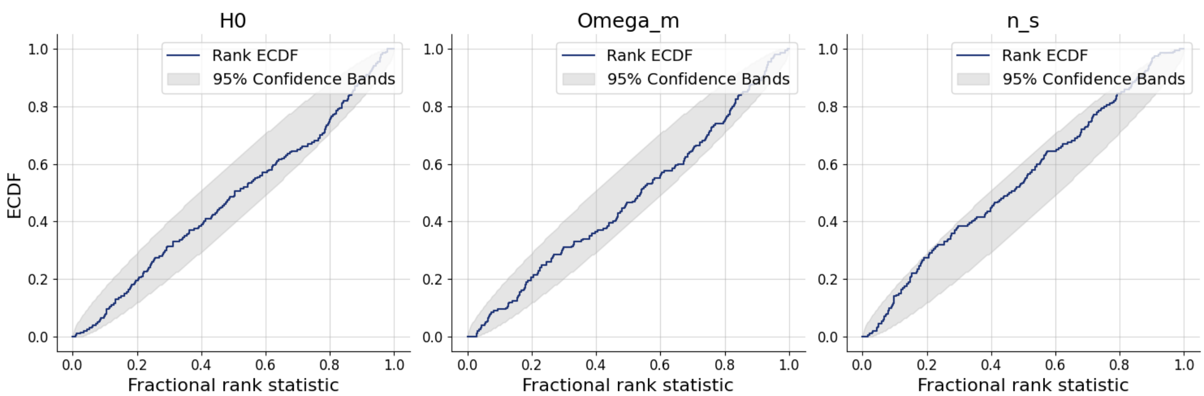
Appendix

```
theta_true = dict(H0=68.0, Omega_m=0.32, n_s=0.965)
obs_pk = simulate_pk(theta_true)["P"].astype("float32")

conds = {"summary_variables": obs_pk[None, :, None]}

posterior = workflow.sample(
    conditions    = conds,
    num_samples   = num_samples,
)

df_samples = workflow.samples_to_data_frame(posterior)
print(df_samples.head(5))

Error in CAMB: must be real number, not tuple
            H0   Omega_m       n_s
0    64.868805  0.212588  0.450235
1    57.345955  0.246956  1.498660
2    58.891125  0.173759  1.388321
3    45.132359  0.135430  1.328867
4    55.748978  0.201402  1.652495
```




H0
Omega_m
n_s
r = 0.650
r = 0.827
r = 0.997
Estimate
Ground truth
Ground truth
Ground truth


H0
Omega_m
n_s
Rank ECDF
95% Confidence Bands
Rank ECDF
95% Confidence Bands
Rank ECDF
95% Confidence Bands
ECDF
Fractional rank statistic
Fractional rank statistic
Fractional rank statistic